

Tutorial 3. Interest Points, Registration and Fitting Solutions

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Interactive lab: cv-tutorial-03-lab.chenrui-fan.chatgpt.site

Source code: github.com/Sosekie/Unibe-CVG-CV-Interactive-Labs

Scope. This tutorial focuses on interest-point detection and description, line fitting, and 2D image registration. Q2 and Q6 extend the lecture with Hessian and Prewitt calculations and are optional. State any additional assumptions you use. The solution sheet repeats each question before its answer.

Interest points

Focus: Harris and Hessian responses; contrast scaling; detector versus descriptor.

Use one-based array coordinates (r, c) , with r increasing down the page and c to the right. The displayed image axes have x to the right and y upward. Define

$$I^{(1)} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}, \quad I^{(2)} = \frac{1}{2}I^{(1)}.$$

The two locations are $p_* = (4, 4)$ and $p_{**} = (6, 4)$. At an image boundary use *reflection without repeating the boundary pixel*: for example, $I(0, c) = I(2, c)$ and $I(7, c) = I(5, c)$; likewise for columns.

Q1. Harris response. Approximate the lecture's Gaussian derivative and Gaussian window with the following unnormalised central first differences and a uniform 3×3 average:

$$I_x(r, c) = I(r, c + 1) - I(r, c - 1), \quad I_y(r, c) = I(r - 1, c) - I(r + 1, c).$$

At each location form the second-moment matrix $M = \frac{1}{9} \sum_{q \in \mathcal{N}_{3 \times 3}} \nabla I(q) \nabla I(q)^\top$ and the score $R = \det M - k(\text{tr } M)^2$ with $k = 0.05$. Compute M and R at p_* and p_{**} in both images. Explain how R changes if every intensity is multiplied by α .

Solution.

For $I^{(1)}$, the 3×3 gradient windows give

$$M_*^{(1)} = \frac{1}{9} \begin{bmatrix} 4 & -1 \\ -1 & 4 \end{bmatrix}, \quad M_{**}^{(1)} = \frac{1}{9} \begin{bmatrix} 6 & 0 \\ 0 & 0 \end{bmatrix}.$$

The negative off-diagonal term follows from the stated upward y -axis. Its sign does not affect the determinant or Harris score. Using $k = 1/20$,

$$R_*^{(1)} = \frac{15}{81} - \frac{1}{20} \left(\frac{8}{9} \right)^2 = \boxed{\frac{59}{405} \approx 0.14568}, \quad R_{**}^{(1)} = 0 - \frac{1}{20} \left(\frac{6}{9} \right)^2 = \boxed{-\frac{1}{45} \approx -0.02222}.$$

If $I \mapsto \alpha I$, then $\nabla I \mapsto \alpha \nabla I$, $M \mapsto \alpha^2 M$ and $R \mapsto \alpha^4 R$. Therefore $M^{(2)} = M^{(1)}/4$ at each point, while

$$R_*^{(2)} = \boxed{\frac{59}{6480} \approx 0.009105}, \quad R_{**}^{(2)} = \boxed{-\frac{1}{720} \approx -0.001389}.$$

The positive response at p_* indicates variation in two directions; the negative response at p_{**} is characteristic of an edge-like neighborhood.

Q2. Hessian response (optional extension). To remove discretisation ambiguity, use the following *direct second differences of intensity*:

$$\begin{aligned} I_{xx}(r, c) &= I(r, c+1) - 2I(r, c) + I(r, c-1), \\ I_{yy}(r, c) &= I(r-1, c) - 2I(r, c) + I(r+1, c), \\ I_{xy}(r, c) &= \frac{1}{4}[I(r-1, c+1) - I(r-1, c-1) \\ &\quad - I(r+1, c+1) + I(r+1, c-1)]. \end{aligned}$$

For $H = \begin{bmatrix} I_{xx} & I_{xy} \\ I_{xy} & I_{yy} \end{bmatrix}$, compute $\det H$ at both locations in both images. What contrast-scaling law does this score obey?

Solution.

In $I^{(1)}$, the derivatives at p_* are $I_{xx} = -1$, $I_{yy} = -1$, and $I_{xy} = -1/4$. At p_{**} they are -1 , 0 , and 0 , respectively; reflection makes the vertical second difference zero at the bottom row. Thus

$$H_*^{(1)} = \begin{bmatrix} -1 & -1/4 \\ -1/4 & -1 \end{bmatrix}, \quad \boxed{\det H_*^{(1)} = \frac{15}{16}}, \quad H_{**}^{(1)} = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \boxed{\det H_{**}^{(1)} = 0}.$$

Second differences are linear in intensity, so $H \mapsto \alpha H$ and $\det H \mapsto \alpha^2 \det H$. Consequently $\det H_*^{(2)} = 15/64$ and $\det H_{**}^{(2)} = 0$. These results use the specified direct second differences; applying the unnormalised first-difference filter twice defines a different discrete operator and gives different numbers.

Q3. Detector and descriptor. Explain the different jobs of an interest-point detector and a feature descriptor when matching two images. Why are scale and orientation often estimated or normalised?

Solution.

A detector selects repeatable image locations, optionally with a scale or orientation. A descriptor summarises the neighborhood of a selected location as a representation that can be compared across images. Matching compares descriptors associated with detections; a descriptor does not itself choose the interest-point locations. Scale and orientation selection or normalisation help the corresponding neighborhoods remain comparable after zoom or image rotation.

Fitting

Focus: ordinary and total least squares; the Prewitt operator as a local plane fit.

Q4. Ordinary least-squares line. For observations (x_i, y_i) , $i = 1, \dots, n$, minimise

$$E(c_1, c_0) = \sum_{i=1}^n (c_1 x_i + c_0 - y_i)^2.$$

Derive the normal equations for $y = c_1 x + c_0$. Under what condition is the fitted slope unique?

Solution.

Setting both partial derivatives of E to zero gives

$$\begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & n \end{bmatrix} \begin{bmatrix} c_1 \\ c_0 \end{bmatrix} = \begin{bmatrix} \sum x_i y_i \\ \sum y_i \end{bmatrix}.$$

Equivalently, with means $\bar{x}, \bar{y}$,

$$c_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad c_0 = \bar{y} - c_1 \bar{x}.$$

The denominator is positive exactly when the x_i are not all equal; this is the condition for a unique slope and intercept in this parameterisation.

Q5. Numerical line fit. Apply Q4 to $(0, -7)$, $(2, -1)$ and $(4, 5)$. Give the fitted line and its sum of squared vertical residuals.

Solution.

Here $\sum x_i^2 = 20$, $\sum x_i = 6$, $\sum x_i y_i = 18$, and $\sum y_i = -3$. Hence

$$\begin{bmatrix} 20 & 6 \\ 6 & 3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_0 \end{bmatrix} = \begin{bmatrix} 18 \\ -3 \end{bmatrix}, \quad \boxed{c_1 = 3, \quad c_0 = -7}.$$

All three points lie on $\boxed{y = 3x - 7}$, so every vertical residual and their squared sum equal zero.

Q6. Prewitt from plane fitting (optional extension). In a 3×3 image patch, let local u increase to the right and local v upward; the top row therefore has $v = +1$. Fit $z(u, v) = au + bv + c$ by ordinary least squares to the nine intensity values z_{uv} at $u, v \in \{-1, 0, 1\}$. Derive a and b and write the corresponding 3×3 masks. Specify whether you are using correlation or convolution.

Solution.

The centred grid has $\sum u = \sum v = \sum uv = 0$, $\sum u^2 = \sum v^2 = 6$ and nine samples. Its normal equations reduce to

$$\text{diag}(6, 6, 9) \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \sum uz_{uv} \\ \sum vz_{uv} \\ \sum z_{uv} \end{bmatrix}.$$

Thus $a = \sum uz_{uv}/6$, $b = \sum vz_{uv}/6$, and $c = \sum z_{uv}/9$. Applied as *correlation* masks in displayed row order, the slope estimates are

$$K_x = \frac{1}{6} \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}, \quad K_y = \frac{1}{6} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}.$$

These are normalised Prewitt masks for the stated upward v convention. Convolution flips each mask by 180° and therefore reverses the sign unless the kernel is adjusted.

Q7. Total least-squares line. Minimise the squared perpendicular residuals

$$E(a, b, d) = \sum_{i=1}^n (ax_i + by_i - d)^2, \quad a^2 + b^2 = 1.$$

Derive the solution using the centred point matrix. Then compute one valid (a, b, d) for the three points in Q5. How does this objective differ from Q4?

Solution.

For fixed (a, b) , differentiation in d gives $d = a\bar{x} + b\bar{y}$. Let U have rows $(x_i - \bar{x}, y_i - \bar{y})$ and let $n = (a, b)^\top$. The remaining objective is $\|Un\|^2 = n^\top U^\top U n$ under $\|n\| = 1$. A unit eigenvector of $U^\top U$ for its smallest eigenvalue minimises it.

For the three points, $(\bar{x}, \bar{y}) = (2, -1)$ and

$$U = \begin{bmatrix} -2 & -6 \\ 0 & 0 \\ 2 & 6 \end{bmatrix}, \quad U^\top U = \begin{bmatrix} 8 & 24 \\ 24 & 72 \end{bmatrix}.$$

One unit null vector is $n = (-3, 1)^\top / \sqrt{10}$, giving

$$(a, b, d) = \left(-\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}}, -\frac{7}{\sqrt{10}} \right).$$

Multiplying all three parameters by -1 describes the same line. Its perpendicular error is zero. Q4 minimises *vertical* errors and requires a finite slope; total least squares minimises *orthogonal* distances and can represent vertical lines.

Registration

Focus: affine maps, point correspondences, and homography estimation.

Q8. Affine transformation. Given corresponding points $(x_i, y_i) \leftrightarrow (x'_i, y'_i)$, write the linear equations for the six parameters of a 2D affine map. State the minimum number of pairs for a unique exact solution and the required geometric condition. What changes when more noisy pairs are available?

Solution.

For

$$\begin{bmatrix} x'_i \\ y'_i \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix},$$

each pair contributes two rows to $A\theta = b$, with $\theta = (a_{11}, a_{12}, a_{21}, a_{22}, t_x, t_y)^\top$:

$$\begin{bmatrix} x_i & y_i & 0 & 0 & 1 & 0 \\ 0 & 0 & x_i & y_i & 0 & 1 \end{bmatrix} \theta = \begin{bmatrix} x'_i \\ y'_i \end{bmatrix}.$$

Three pairs provide six equations. The three source points must be non-collinear so that A has rank six; three merely distinct points may be collinear and insufficient. With additional noisy pairs, solve a full-rank overdetermined system by least squares rather than demanding exact agreement.

Q9. Affine numerical example. Estimate the map from $(0, 0) \mapsto (1, -2)$, $(1, 0) \mapsto (3, -3)$ and $(0, 1) \mapsto (2, 1)$. Predict the image of $(1, 1)$.

Solution.

The first pair gives $(t_x, t_y) = (1, -2)$. Comparing the second pair with the first gives the first column $(a_{11}, a_{21}) = (2, -1)$; comparing the third pair gives the second column $(a_{12}, a_{22}) = (1, 3)$. Therefore

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad (1, 1) \mapsto (4, 0).$$

Q10. Homography. For homogeneous points $p_i = (x_i, y_i, 1)^\top$ and $p'_i = (x'_i, y'_i, 1)^\top$, a nonsingular homography satisfies $p'_i \sim H p_i$. Eliminate the unknown per-point scale to obtain two homogeneous linear equations per pair in the nine entries of H . Explain how to estimate H with SVD when correspondences are noisy. State the minimum number of pairs, the nondegeneracy condition, and the scale ambiguity.

Solution.

Write $h = (h_{11}, h_{12}, h_{13}, h_{21}, h_{22}, h_{23}, h_{31}, h_{32}, h_{33})^\top$. Since the third coordinate of $H p_i$ equals $h_{31}x_i + h_{32}y_i + h_{33}$, the first two coordinates give

$$\begin{bmatrix} x_i & y_i & 1 & 0 & 0 & 0 & -x'_i x_i & -x'_i y_i & -x'_i \\ 0 & 0 & 0 & x_i & y_i & 1 & -y'_i x_i & -y'_i y_i & -y'_i \end{bmatrix} h = 0.$$

Stack these blocks into $Ah = 0$. With noisy data, minimise $\|Ah\|_2$ subject to $\|h\|_2 = 1$ and reshape the right singular vector of A for its smallest singular value into H ; normalising image coordinates before DLT improves numerical conditioning. A homography has eight degrees of freedom because H and βH represent the same map for every $\beta \neq 0$. Thus at least four point pairs are needed, with the points in general position (in particular, no three of the four source points collinear, and likewise for the target points). For four exact nondegenerate pairs, A has rank eight and determines H up to scale. Fixing $h_{33} = 1$ is only a possible gauge when that entry is nonzero; it is not a universal normalisation.

Q11. RANSAC for registration. Putative point correspondences include outliers. Explain how RANSAC estimates a homography from them and why fitting once to all matches may fail. If the inlier fraction is $w = 0.5$, the minimal sample contains four matches, and the desired probability of drawing at least one all-inlier sample is $p = 0.99$, how many independent random samples are needed? Use the standard approximation that each selected match is independently an inlier, and assume independent nondegenerate samples.

Solution.

Repeat: draw four matches, reject degenerate configurations, fit a candidate homography, and count matches whose reprojection error is below a chosen threshold. Keep the best consensus set, then refit using its inliers (and, if appropriate, refine the reprojection error). A single least-squares fit to all matches can be badly distorted by outliers because squared residuals give large errors high weight.

One random sample is all inliers with probability w^4 . The probability that none of N independent samples is all inliers is $(1 - w^4)^N$. Require $(1 - w^4)^N \leq 1 - p$, so

$$N \geq \frac{\log(1 - p)}{\log(1 - w^4)} = \frac{\log(0.01)}{\log(15/16)} \approx 71.36.$$

Hence $\boxed{N = 72}$ samples suffice under the stated assumptions. The estimate depends on the actual inlier fraction and on valid random samples; it is not a guarantee when these assumptions fail.